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Task Description:	Osteoporosis, the progressive loss of bone density and strength which cripples tens of millions on our planet, distinguishes itself as perhaps the greatest physiologic obstacle to an extended human presence in space. The principal objectives of this proposal are to establish the efficacy of a unique, biomechanical countermeasure to inhibit bone loss in an animal model of disuse osteoporosis, and correlate this regulatory influence to the expression patterns of several genes critical to bone formation and resorption. Using a ground based model of microgravity, the tail-suspended rat, we have shown that brief exposure (10 minutes) to extremely low magnitude (0.25g, engendering < 5 microstrain), high frequency (30-90 Hz) mechanical signals will inhibit the bone loss which typically parallels disuse, even though 10 minutes of full weightbearing failed to curb this loss. Longer-term experiments in sheep have shown this stimulus to be strongly anabolic, increasing bone mineral density, trabecular number and connectivity, and improving bone strength. In a series of four specific aims, we will use several morphometric assays on the mouse model of tail-suspension to		

Task Description:	rigorously establish the efficacy of a specific mechanical signal (10 minutes at 30Hz, 0.3g; parameters being used in clinical trials to inhibit bone loss in the elderly) to inhibit and/or reverse 28 days of disuse osteopenia. In an effort to understand the mechanisms by which this signal is anabolic, we will also monitor the temporal and spatial expression of nine genes, each indicative of a specific process of bone formation or resorption. The use of the mouse will facilitate many aspects of the protocol, including comprehensive genomic profiling and expedited access to spaceflight. Considering that many flight opportunities are brief and thus do not permit long term morphologic adaptations in bone to occur, combining the molecular with the tissue level strategies will facilitate establishing countermeasure efficacy even following short term exposure to microgravity. In essence, this work represents a critical step in establishing a physiologically based, non-pharmacologic, non-invasive treatment for osteoporosis, for use on earth or in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understand how gravity influences the quantity and quality of bone, and how mechanical information may be used to prevent and/or reverse osteoporosis, a disease which effects over 20 million americans here on earth

Summary: Following one year of support, preliminary data have been collected from bedrest subjects (control), as the first stage in evaluating efficacy of a unique, non-pharmacologic countermeasure for disuse osteoporosis. This low magnitude (0.3g), high frequency (30Hz) mechanical signal has been shown to be anabolic to bone. The means of delivering this mechanical information to the supine subject are described within, as well as details of the assays to evaluate efficacy. Work will continue in the second year to collect baseline, 60d and 90d data from controls and subjects subject to 10minutes per day of the vibration stimulus. Introduction:
The osteoporosis which develops in microgravity is one of the greatest hurdles to an extended human presence in space. Earth-based animal and human studies have demonstrated that extremely low magnitude mechanical stimuli (LMMS), if imposed at a high frequency, is strongly anabolic to the skeleton, and can serve to inhibit the bone loss, which typically parallels disuse. This experiment is designed to evaluate the efficacy of this unique biomechanical countermeasure to inhibit the disuse induced osteoporosis seen in long term bed-rest, the closest ground based equivalent of microgravity. To achieve this in a non-invasive, non-pharmacologic means would have tremendous impact not only in space, but would also address the bone loss which plagues over 20 million people world wide each year on earth.

Project Aims: 1) Show that application of low magnitude (0.3g), high frequency (30Hz) mechanical stimulation, will reduce the loss of bone seen with long term disuse 2) Show that application of low magnitude, high frequency mechanical stimulation will improve the postural control of subjects undergoing long term bed-rest 3) Determine if long term bed-rest affects the sensitivity of the lower extremities. 3.b) Determine if the application of low magnitude, high frequency mechanical stimulation will inhibit the changes if they exist

The first aim of the project applies to the following CPR questions. • 1.d: What biophysical modalities, nutritional modification, and pharmacological agents will most effectively minimize the decrease in bone mass due to extended hypogravity exposure? • 1.e: What are the specifics of the optimal exercise regimen with regard to mode, duration, intensity and frequency, to be followed during exposure to hypogravity to minimize the decrease in bone mass? • 1.f: What combination of exercise, biophysical modalities, nutritional modifications, and/or pharmacological agents is most effective, efficient, and safe in preventing bone loss in hypogravity?

By showing a reduction in the loss of bone with daily treatment of mechanical stimulation, we will be providing a treatment for the loss of bone seen in long term space flight. Analysis of Bone Mineral Density (BMD) will be performed on control and experimental groups using qualitative ultrasound, DXA, and CT scans.

The second aim applies to the following CPR questions • 14g: What in-flight training techniques (e.g., virtual reality simulations, treadmill with vibration isolation system, artificial gravity) can be used to alleviate the risks of impaired balance control and movement coordination as astronauts land and (re)adapt to Earth, Moon, or Mars gravity? • 14j: What artificial gravity exposure regimens (g-level, angular velocity, duration, and repetition) will ameliorate the physiological and vestibular deconditioning associated with hypogravity during transit phases of a mission in order to increase the capability to perform operational tasks after landing and throughout rehabilitation

Analysis of the postural control of the subjects will be measured using a Kistler force plate. It is hypothesized that daily treatment of mechanical stimulation will improve the postural control of subjects. Long term disuse, like hypogravity, causes a decrease in postural control which results in an increased risk in falling. By showing an improvement in the parameters indicative of good postural control in the subjects who undergo the treatment over the control group, we will be able to show that the treatment is an effective means to reduce the risk of falling upon returning to a gravity environment, as well as an improved mobility.

The third aim applies to the following CPR questions • 13k: What artificial gravity regiments will ameliorate the physiological and vestibular deconditioning associated with hypogravity during transit phase of a mission in order to increase the capability to perform operational tasks during flight, entry, and landing? • 14a: What are the physiological bases for disruption of balance, locomotion, and eye-hand coordination following g-phase transitions?

The effect of hypogravity on the sensitivity of the feet and ankles has not been studied extensively. Most study on physiological changes associated with balance and locomotion have centered on the vestibular system. However, a significant portion of postural control is contributed by the nerves of the feet and ankles. Sensation of plantar pressure and torque around the joints gives us information that we use in balance. If these sensations are impaired from disuse, then our balance is impaired as well. We will be using an array of tests to determine if any of these parameters are affected by long-term bed-rest, and if our daily treatment can affect the parameters.

A principal goal of the National Aeronautics and Space Administration is to implement long-term human exploration of space. Whether this is in the near future, by habitation of the International Space Station, or within the next few decades by settling a permanent manned Moon base and embarking on a mission to Mars, it is clear that we, as a people, must develop a better understanding of the ability of humans to tolerate an extended exposure to microgravity. As defined by the National Research Council report entitled: A Strategy for Research in Space Biology and Medicine in the New Century, there are several areas of fundamental scientific investigation which must be addressed to make long-term space exploration not only feasible, but safe. Summing up their recommendations, which echo those made by previous reports published by the National Research Council's Space Studies Board, the principal physiologic hurdle to man's extended presence in space is the osteopenia which parallels reduced gravity. The magnitude and rate of the loss is

staggering; astronauts lose bone mineral in the lower appendicular skeleton at a rate approaching 1.6% per month. Thus, in a 2.5-year return-trip to Mars, half of an astronaut's bone density may vanish, severely jeopardizing their health and well-being. Addressing this concern, NASA has developed a Critical Path Roadmap (CPR; 2001) to define its bioastronautics research priorities towards reducing or eliminating the risks to astronaut health and safety during and after spaceflight. This CPR emphasizes musculo-skeletal deterioration in general, and osteoporosis in particular, as research areas of highest priority. The most significant consequences of spaceflight may be realized only upon return to planetary gravitational fields. To compound this problem, it is apparent that full recovery of bone mass may never occur, predisposing the astronaut to skeletal complications later in life. Considering the similarities between space and aging induced bone loss, an indisputable benefit of such a countermeasure would be its potential as a treatment for the osteoporosis which plagues 25 million people in the U.S. (NIH Consensus Development Conference, 2000). Unfortunately, the great majority of flights into space are brief (e.g., 7d COSMOS 1667 & Spacelab 3; 14d COSMOS 2044 & STS 54), making it difficult to identify the morphologic consequences of the removal of gravity on skeletal morphology, or the efficacy of any given countermeasure. It must be added that the limited numbers of astronauts and payload specialists, the variability of factors which influence bone mass and morphology (e.g., age, gender, body mass index, nutritional status, exercise habits), and the range of space mission durations, have made it extremely difficult to determine the efficacy of any given countermeasure. By using a ground-based bed-rest study as a simulation of the disuse of the axial skeleton seen in space flight, we are able to perform a more controlled experiment with a larger subject group, without using the very critical time of astronauts in space. Therapies that increase bone formation are highly desirable, but unfortunately, not nearly as well developed as antiresorptive interventions. Anabolic agents currently under investigation, including parathyroid hormone (PTH), fluoride, and insulin-like-growth factor-I (IGF-I), have important and significant disadvantages. For example, PTH must be administered by subcutaneous injection, may cause hypercalcemia and hypercalciuria, and concerns about osteosarcoma persist. Fluoride may cause gastrointestinal side effects and there is evidence of increased skeletal fragility and microfractures with increasing doses. IGF-I administration is associated with hypoglycemia, and has not been well studied as a therapeutic agent for bone. Even dismissing these concerns, such pharmacologic interventions seem inappropriate for long-term spaceflight, as they – in essence – ignore the etiologic basis for the disease, the removal of gravity, and may place otherwise healthy astronauts at undue risk. Further, the administration of a drug, as well as its pharmacokinetics, may be hampered in microgravity. With the removal of gravity considered the basis for the osteopenia, it is reasonable to examine the potential of mechanical stimuli (e.g., exercise) as a countermeasure.

Materials & Methods:

Overview This study is aimed at showing that LMMS can reduce the physiological changes seen in bone and postural control during long term bed rest. To perform these tests, control subjects undergoing 90 days of bed-rest will be tested at 0, 60, and 90 days of bed rest. Control groups will undergo mechanical stimulation for 10 minutes a day, 5 days a week for the duration of the study, with measurements taken at the same time points

Application of Stimulation Stimulation is performed using an oscillating platform. The platform is designed to vibrate at a rate of 30 Hz, at an amplitude of 0.3 g, with doses of ten minutes each day. Typically, the subject would stand on the platform, which is roughly the size and shape of a bathroom scale, for their daily treatment. However, in bed-rest as in space, the subjects do not have gravity pulling them to the plate. To overcome this, the plate will be hung on its side from a moving platform. This platform can be moved from room to room. The subject will put on a padded vest, which will then attach to the plate via two springs, one at each side of the plate. The springs will apply a force to the subject's shoulders, coupling them to the plate. The magnitude of the applied force will be equal to 60% of the subject's body weight. In previous work we found that at 60% body weight, a subject will undergo stimulation of 70-80% of the magnitude seen when standing on the plate under the effects of gravity. Balancing subject comfort and signal magnitude was the rational for settling at this number.

Postural Control Analysis of postural control is performed using a Kistler force plate. The force plate contains four piezoelectric load cells. The plate can detect forces in the vertical and both horizontal directions, as well as all three moments. When a subject stands on the plate, small shifts in posture will result in changing forces applied to the plate. The location and magnitude of the resultant of these applied forces is recorded and defined as the Center of Pressure (COP). The subject stands barefoot on the plate with their hands at their sides. Foot placement is defined by a set of markings on the surface of the force plate, with the feet slightly spread and splayed outward at 10 degrees (Figure 1). The subject closes their eyes, and stand as still as they can with their head forward for 4 minutes. After 5 minutes of rest, the measurement is repeated with the subject's eyes open. A marker is placed on a wall 2 meters from them at eye level, and they are instructed to focus on the marker for the duration of the measurement. The force plate output will be recorded at 50 Hz using the Kistler Bioware software. The software will be used to convert the forces recorded by the force plate to COP.

Nerve sensitivity The sensitivity of the foot is related to the balance of an individual. Those with diminished sensitivity have an increased risk of falling. To determine if the daily applications of LMMS will have an effect on the sensitivity of the feet and ankles, the following tests will be performed.

a. Monofilament light touch sensitivity The first test that is performed is a light touch sensitivity test. A WEST-foot monofilament set will be used to perform this test. The WEST-foot monofilament device consists of a pen sized stick with 5 plastic monofilaments attached perpendicularly, which can be rotated to select a single filament (Figure 2). The filaments are each of a different thickness. This test is performed with the subject lying in bed. The subject's non-dominant foot is used. For the test, they will be barefoot and have their eyes closed. The tester then lightly presses the filament against the skin on the foot of the subject, just enough so that a small indentation is made. The subject states where they feel the touch. The tester will not prompt the subject, if they don't feel the touch, then it is counted as a failed identification. If the subject names an incorrect location, it is counted as a failure. The test is done randomly across different locations of the foot including the big toe, small toe, heal, ankle, and metatarsal heads. Before the test is performed, each location is touched with the thickest filament with the subject's eyes open, so they know what sensation to expect. The test is performed with all 5 filaments, starting with the thickest, and working down.

b. Two Point Discrimination The second test that is performed is the two point discrimination test. To perform this test, we use the Touch-Test Two-Point Discriminator (Figure 3). This test is designed to determine the smallest distance that a subject can identify as two separate points. As with the monofilament test, this is performed first on the subject with their eyes open, so the subject is aware of how to perform the test. The test is performed again with the subject's foot bare and eyes closed. A random gap will be chosen first. The skin of the subject's foot is lightly touched with the chosen size, until a slight indent in the skin is made. The subject then states if they feel one or two points. If they detect one but

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two points are used, the next largest gap is chosen. If they detect two, the next smallest is chosen. This is repeated until the smallest gap that they can correctly detect 50% of the time is found. The next larger gap is given as the smallest detectable distance. This test is performed at 5 locations on the foot: the large toe, small toe, heel, first metatarsal head, and fourth metatarsal head.

c. Joint Pronation The third test performed is a joint pronation test. The tester grips the subject's large toe gently at the first joint, with one finger on each side. The tester grips the subject's ankle to keep it from bending. The subject is told to relax, so that the tester can freely move the toe joint. When the subject has loosened their muscles, the tester will bend the toe joint in small, 5 degree angles. The subject states if they felt the toe move up or down. The toe is moved rapidly up and down at random in the same small angle changes 15 times. The number of incorrect answers is counted. If the subject delays more than a second before answering, it is counted as incorrect. This test is also performed with the ankle joint of the subjects if they fail more than 30% of the time at the toe. In this case, the tester will steady the leg of the subject at mid-calf. He then grips the foot of the subject at the boney processes at each side of the foot below the large and small toe. The test protocols are identical, except that the ankle joint is rotated up or down at 5 degree intervals instead of the toe. For both cases, all motion is kept close to the middle range of motion of the joint, the joint will not be hyper or hypo extended.

d. Vibration Sensitivity The fourth sensitivity test performed is a vibration sensation exam. Two tuning forks are used to perform this test. One fork has a natural frequency of 30 Hz, the second at 256 Hz. The tester strikes the fork to make it vibrate, while holding it at its neck. The base of the neck is held against the boney process below the large toe of the subject's foot (Figure 4). This is repeated with the second fork, so that the subject has felt each with their eyes open. When this is done, the subject closes their eyes. The subject will then be stimulated with each fork five times with their eyes closed. The order of the forks will be random. The subject will be asked if it is the high or low frequency fork. The number of correct choices will be recorded. This will be repeated at the ankle as well.

Bone Mineral Density Quantitative Ultrasound Quantitative Ultrasound measures the properties of bone by passing an ultrasound signal through the bone. Measurements are performed on the non-dominant ankle of each subject. Depending on the density and structure of the bone, the magnitude of the signal that will be attenuated will change. Also, the velocity of the signal which exits the bone will be affected as well. By analyzing these two parameters, information on the quality of the bone can be ascertained. Measurements are performed using an ultrasound scanning device designed and built in our laboratory. The subject sits with their foot in a water bath to help couple the ultrasound signal (figure 5). Ultrasound transducers raster to scan the region of interest. CT Scans BMD of the lumbar spine (L1-4), right and left proximal femur (neck, trochanter, and Wards triangle, total proximal femur), and non-dominant radius (distal third) will be measured by CT scan. Scans are performed by staff at the bed-rest facility in Galveston, TX. The method and location of the scans are designed to match those performed are part of the standard routine of examination of astronauts returning from space.

Results: Measurements have been made for 4 control test subjects at day 0. Day 60 measurements will be made during the first week of May, 2005.

Application of LMMS We've determined that application of mechanical stimulation can be achieved in a supine individual. By attaching a supine subject to the vibrating plate through a harness and spring system, coupling between the feet of the subject and the top surface of the plate is achieved. As the load applied to the shoulders of the subject increases, the magnitude of stimulation the subject undergoes increases. This was determined by attaching an accelerometer to a bite-bar to measure the acceleration of the subjects head. By comparing this to the acceleration of the surface of the plate, the percent transmission of the signal was calculated (Figure 6). With 60% of the subject's body weight applied to their shoulders, the %transmission is approximately 70-80% of that of a subject standing on the plate under the effects of gravity. Figure 6: Relationship between the load applied to a supine subject and the transmission of vibration. Data of 10 subjects shown.

Foot sensitivity

The tests of the sensitivity of the foot showed that each subject had a typical sensitivity of a healthy individual (table 1). Comparisons will be made with those after extended bed-rest to determine if disuse causes a change in sensitivity. Big Toe Small Toe 1st metatarsal 4th metatarsal heel ankle Joint Pronation@ 97±7% mean +/- SD Two Point Discrimination# 8.5±2.4 11.25±4.9 16±2.7 9.5±2.6 17.25±5.5 N/A West-foot Monofilament* 4.25±0.5 4.5±0.6 4.5±1 4.75±0.5 4.25±0.96 3.25±1.7 Vibration Sensitivity@ 79±16% mean +/- SD @ % correct ± SD * number correct ± SD # mean ± SD Table 1: Baseline sensitivity scores of control subject group

Bone Density

Scans have been performed on the non-dominant ankle by quantitative ultrasound. Broadband Ultrasound Attenuation (BUA) and Ultrasound Velocity (UV) are calculated from the raw data. Both are indicators of the structure and composition of the bone (figure 7). The more dense the bone is, the more signal will be attenuated and slower the velocity will be. Thinner, less structured bone will attenuate signal less, and have an overall greater velocity.

Figure 7: Broadband Ultrasound Attenuation (left) and Ultrasound Velocity (right). Data of four control subjects shown as mean +/- SD.

DEXA and CT data of the day 0 control group has been collected but not fully analyzed. Figure 8 shows two images of the scans of one subject. The first image shows the proximal tibia. The second image shows the lower lumbar spine of the same subject.

Postural Control

Postural measurements at day zero were made on four control subjects. Postural data consists of Center of Pressure (CoP) information collected using a Kistler force plate. CoP is defined as the location of the ground reaction force under the feet of the subject. Figure 9 shows 4 minutes of CoP measurement for one of the subjects with their eyes closed. It can be seen that the subject sways mostly in the anterior/posterior direction. From the CoP recording, measurements of ML and AP sway magnitude and velocity, total path length, and frequency composition can be made. These numbers give us information on the nature of the postural control of the subject. A subject with a larger sway magnitude and velocity will be a greater risk of falling than a subject who sways less. Day 0 measurements show all control subjects have postural control of a typical healthy adult (figure 10). Figure 10: Left: Root mean square sway magnitude (m) in

the AP (blue) and ML (purple) directions for the four controls subjects. Right: RMS sway velocity (m/s) in the AP (blue) and ML (purple) directions.

Discussion: • 1.d: What biophysical modalities, nutritional modification, and pharmacological agents will most effectively minimize the decrease in bone mass due to extended hypogravity exposure? • 1.e: What are the specifics of the optimal exercise regimen with regard to mode, duration, intensity and frequency, to be followed during exposure to hypogravity to minimize the decrease in bone mass? • 1.f: What combination of exercise, biophysical modalities, nutritional modifications, and/or pharmacological agents is most effective, efficient, and safe in preventing bone loss in hypogravity?

Our data so far shows baseline data for four healthy controls. Upon completion of the bed-rest study, we will be able to compare the bone mineral density of control subjects who under went 90 days of bed-rest simulating microgravity, and an experimental group who underwent the same bed-rest, but had daily treatments of 10 minutes of mechanical stimulation. If the data collected for bed-rest mirrors what we have seen in post-menopausal women as we hypothesize, there will be a significant drop in the loss of bone due to disuse. If this is the case, it may be possible to effectively reduce and/or stop the loss of bone seen in extended hypogravity. At present, exercise regimes in the ISS and shuttle have not been effective in decreasing the loss of bone in space. LMMS has been shown in multiple ground based models to reduce and/or prevent the loss of bone in at risk groups. Animal studies, as well as human studies of post-menopausal women and children with muscular dystrophy have shown significant improvement in bone density when compared to controls. By loading the experimental subjects with 60% of their body weight, we will be creating a short term load on their axial skeleton, which will be stimulated with the mechanical vibrations generated by the plate. Because the treatment only requires the astronaut to stand upright, they are free to perform other tasks which do not require extensive motion while they undergo treatment, such as reading, writing, or computer work. The plate itself isn't much larger than a bathroom scale. Considering the confined volume of the ISS and any space vehicle, and the high value of crew time, the treatment method is an improvement over time intensive exercise regimes requiring large equipment. • 13k: What artificial gravity regiments will ameliorate the physiological and vestibular deconditioning associated with hypogravity during transit phase of a mission in order to increase the capability to perform operational tasks during flight, entry, and landing? • 14a: What are the physiological bases for disruption of balance, locomotion, and eye-hand coordination following g-phase transitions?

We are investigating whether there is a loss in sensitivity in the feet and ankles due to extended disuse. Foot and ankle nerve sensitivity is a major contributor to postural control. If there is a loss of sensation of the feet due to disuse and/or hypogravity, then specific training can be targeted to help alleviate the loss of sensation to help reduce the impaired balance control of astronauts re-adapting to gravity If there is a significant change in sensitivity, we will also be able to determine if the daily treatment for bone mass has an effect on nerve sensation as well. It is possible that the daily stimulation of the treatment may help alleviate the loss of sensation from disuse. • 14g: What in-flight training techniques (e.g., virtual reality simulations, treadmill with vibration isolation system, artificial gravity) can be used to alleviate the risks of impaired balance control and movement coordination as astronauts land and (re)adapt to Earth, Moon, or Mars gravity? • 14j: What artificial gravity exposure regimens (g-level, angular velocity, duration, and repetition) will ameliorate the physiological and vestibular deconditioning associated with hypogravity during transit phases of a mission in order to increase the capability to perform operational tasks after landing and throughout rehabilitation

Astronauts returning to earth have a significantly impaired balance control system, which over time re-adapts to gravity. However, due to the loss of bone mass from extended hypogravity, the risk of fracture is elevated. We aim to show that the daily use of mechanical stimulation will reduce the changes in indicators of postural control from extended disuse. Data collected at baseline show that the control group have healthy normal postural control. After 60 and 90 days of bed-rest, we expect these values to worsen dramatically. Studies by other groups have shown that high frequency mechanical stimulation can help improve balance control in stroke patients. Although the exact mechanism is unknown, we are seeking to determine if our regimen of daily vibration treatment will improve the postural control after disuse when compared to controls. If the impairment of balance control in astronauts can be reduced with out treatment, there are multiple benefits. Firstly, the astronauts will be more able to safely and effectively complete tasks upon re-entry to a gravity environment. Secondly, if their postural control is improved, they are at a lower risk of falling. Considering the loss of bone seen in long term hypogravity, a reduction in the risk of falling will result in a reduction in the risk of bone fracture. If the treatment is also successful in the treatment of the loss of bone, then the risk of bone fracture from extended hypogravity will be reduced even more significantly.

OTHER INFORMATION AND MATERIALS: - A complete listing of all presentations this grant supports: A low intensity mechanical countermeasure to prohibit osteoporosis in astronauts during long-term spaceflight. C. Rubin, B. Adler, J. Muir, B. Lee, Y-X Qin, and S. Judex. Bioastronautics Working Group, Galveston, Tx. December, 2004.

- A complete listing of all published abstracts--as above.

- A complete listing of all published articles none.

- A complete listing of all other published materials (books, chapters, newspaper releases) none.

- A complete listing of Undergraduate, Graduate, and Post doc's that were supported Graduate Students and Post Docs Supported: Jesse Muir, Grad Student; Beney Lee, Grad Student; Ben Adler, Research Assistant.

Bibliography Type: Description: (Last Updated: 10/22/2009)